
collaboration

42 messages

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Mon, Dec 29, 2025 at 6:12 AM

Dear Swadesh,

Hope all is well with you.

I have a graduate student working on the PFFDTD code to study the plasma shielding of an antenna in space.

We are trying to reproduce some of the results in this paper (<https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2008ja013097>). It appears to me that for such problems, it is better to apply current to the antenna rather than voltage.

I wanted to know your opinion on this. Would you please help us in this regard?

Regards,
Alireza

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Tue, Dec 30, 2025 at 10:24 AM

Dear Alireza,

Good to hear from you. I hope you are doing well too.

I need to think about this problem a bit more carefully, but I am happy to collaborate.

I will look into why voltage excitation was originally chosen by Jeff Ward when creating the PFFDTD code and research how current excitation differs according to relevant references. The Tu et al. (2008) explicitly model a system driven by a high-voltage potential to investigate electrostatic charging phenomena. The PF-FDTD is also voltage driven. In Jeff's thesis, he found that the current distribution is not triangular at the resonances. Since sheath is a voltage-driven phenomenon, I suspect we would be better off sticking to voltage driving. I will get back to you once I have more clarity on this.

Best regards,

Swadesh Patra

[Quoted text hidden]

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Swadesh

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Tue, Dec 30, 2025 at 9:19 PM

Dear Swadesh,

The reason for the idea to change voltage source to current source is that our initial simulation results do not show the bias due to faster electron charge onto the probe (please see the figure 2 in attached paper).

I think ChatGPT can be employed to make the required changes in the code. Unfortunately, I don't have access to AI.

Regards,
Alireza

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2 attachments**Journal of Geophysical Research Space Physics - 2008 - Tu - Plasma sheath structures around a radio frequency antenna.pdf**

471K

 **collision-3_merged.pdf**
2621K

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Wed, Dec 31, 2025 at 12:01 PM

Dear Alireza,

Thank you for the clarification and for sharing the paper.

Do you have an updated version of the code on GitHub that we can use? I have access to AI tools and can experiment with the code to see if we can produce the desired results.

Regards,

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Wed, Dec 31, 2025 at 12:13 PM

Dear Swadesh,

Attached please find a copy of the code that we are using.

Regards,
Alireza

[Quoted text hidden]

 **pffdttd.rar**
15K

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Wed, Dec 31, 2025 at 1:02 PM

Dear Alireza,

Thank you for sending the code.

I will try to run the script to reproduce your current results and will start working on the modifications shortly. It will take me some time to go through everything, so please let me know if there are any specific project deadlines I should be aware of.

Best regards,

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Wed, Dec 31, 2025 at 1:08 PM

Dear Swadesh,

I appreciate your help. It would be great if you could please update me by January 15 so my student can finish the work. We plan to investigate the pump frequency with respect to plasma frequency and gyro frequency, Background

magmetic field orientation, collision frequency, and plasma temperature, as well as antenna structure on plasma sheilding in lower and upper ionosphere.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Jan 24, 2026 at 8:32 PM

Dear Swadesh

Hope all is well with you.
I was wondering if you could make any progress.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Sun, Jan 25, 2026 at 1:56 PM

Dear Alireza,

I have made some progress on the code, and I apologize for the delayed response. An organizational shift at work required me to prioritize other tasks recently.

I will send you more detailed updates in a couple of days.

Best regards,

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Thu, Jan 29, 2026 at 8:19 AM

Thank you very much for the update. I look forward to hearing from you.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Feb 7, 2026 at 12:01 AM

Dear Swadesh,

Hope all is well with you.
I was wondering if you could make progress on the code.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Feb 9, 2026 at 12:14 AM

Dear Alireza,

I have managed to speed up the code execution using AI, and it now runs significantly faster. You can find the updated repository here: sunswadesh/Plasma-Fluid-Simulation-PFFDTD (<https://github.com/sunswadesh/Plasma-Fluid-Simulation-PFFDTD>).

Please look for pffdt_parallel.exe and pffdt_parallelMP.exe. The *MP.exe version should be the fastest of the three executables. You can compile them by running the compile_manual.bat file from the main project folder. I have added some python visualization scripts as well.

I am still working on the sheath aspects. I am exploring a few ideas, but I do not have satisfactory results to share yet. I will keep you posted, hopefully in the next few days we get some positive results.

Regards,

Swadesh Patra

[Quoted text hidden]

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Swadesh

Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Feb 9, 2026 at 12:18 AM

To: swadesh patra <swadeshp@gmail.com>

Dear Swadesh,

I very much appreciate your feedback. I will let you know if I had any problem running the code,

Regards,

Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Feb 9, 2026 at 7:55 AM

To: swadesh patra <swadeshp@gmail.com>

Dear Swadesh,

We could run the code. It is actually much faster. The only issue is that it does not dump velocity and density in the output file.

Regards,

Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>

Mon, Feb 9, 2026 at 10:14 AM

To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

I will review it tonight and get back to you. I might have broken something in the last updates. I should run the regression tests after any major updates.

You have got me excited and with AI it's a lot of fun to add features 😊.

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Feb 9, 2026 at 10:22 AM

To: swadeshp@gmail.com



Alireza reacted via [Gmail](#)

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Mon, Feb 9, 2026 at 10:52 AM

Dear Swadesh,

It is actually easy to make the code dump the output data as mentioned below. I hope to hear good news from you soon about the sheath.

In this file (pffdttd.cpp), line 151, plasma=0 should be changed to plasma=1

Regards,
Alireza

On Mon, 9 Feb 2026, 20:45 swadesh patra, <swadeshp@gmail.com> wrote:

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sun, Feb 22, 2026 at 11:54 PM

Dear Swadesh

I was wondering if you could please update me on sheath simulation.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Feb 23, 2026 at 9:44 AM

Dear Alireza,

I apologize for the delay in my response. I have been quite burdened with work recently, which has limited my time to focus on the simulation.

I expect to have more time next weekend to dedicate to the sheath simulation and will provide you with an update then.

Best regards,

Swadesh Patra

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Mon, Feb 23, 2026 at 10:21 AM

I very much appreciate your help.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Apr 11, 2026 at 12:20 AM

Dear Swadesh,

Hope all is well with you.

I was wondering if you could make progress. Unfortunately, my student is waiting for the code from you to complete her thesis regarding plasma sheath simulation.

I appreciate your help.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Sat, Apr 11, 2026 at 6:33 AM

Hi Alireza,

I am happy to hear back from you. I hope all of you are safe and well. We keep following the terrible news coming from Iran. I assumed the work to be the last thing on your mind in these times and put the work on lower priority. I will push some updates soon

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Apr 11, 2026 at 11:39 PM

Dear Swadesh,

I appreciate your email. Thank God, my family, and I are still alive and safe. I appreciate your help.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Tue, Apr 21, 2026 at 5:16 AM

Dear Swadesh,

Hope all is well with you. I was wondering if you could make any progress.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Tue, Apr 21, 2026 at 2:45 PM

Hi Alireza,

We went on a vacation so unfortunately couldn't make much progress. We will be back next weekend. I hope to share the updates then. Apologies again for the delay.

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Tue, Apr 28, 2026 at 9:11 PM

Deas Swadesh,

I was wondering if you could please give me feedback on your progress.

We are also working to simulate the effect of plasma clouds on radio wave characteristics (amplitude, phase, modulation, propagation angle) as the plasma cloud parameter changes. We have an issue that the far field data from the simulation is not reliable. In fact, we need to increase the box size so much that the computation source does not support it. I was wondering if you could please help us to generate the plane wave propagating in the code (no physical antenna to be included) so we can just simulate the impact of plasma clouds on plane wave characteristics.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Thu, Apr 30, 2026 at 11:10 AM

Dear Alireza,

I haven't been able to work on it much since getting back from vacation, as I've been stuck with some additional work at my "paid job" 😊. I am optimistic about getting some time in the next few days to focus on this.

Regarding the far-field issue, the calculations in the current version would indeed make the box too big. It's possible there are issues with the absorbing boundary conditions that we need to look into.

I'll keep you posted.

Best regards,

Swadesh Patra

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, May 2, 2026 at 4:06 AM

,Dear Swadesh
I appreciate your help and time in advance.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, May 9, 2026 at 11:45 PM

Dear Swadesh

I hope you can give me feedback on the sheath simulation.

Regards,

Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Sat, May 16, 2026 at 8:49 PM

Dear Alireza,

I apologize again for the delay in my response. I have been working on the simulation with considerable help from AI, and I usually push all my trials to the repository for record.

I am still working to get meaningful results for the sheath simulation, but I feel I am getting close. I will let you know as soon as I have something I am happy with. Regarding the far-field model, that is also currently under parallel planning.

I wanted to keep you in the loop and thank you for your continued patience. I hope to share good news with you soon.

Best regards,

Swadesh Patra

[Quoted text hidden]

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Swadesh

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, May 16, 2026 at 10:23 PM

Dear Swadesh,

Thanks a lot. That is good news. For the far-field issue, we could resolve the problem. I will keep you posted.
I look forward to hearing from you.

Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Jun 6, 2026 at 2:25 AM

Dear Swadesh,

Sorry to bother you again.
I was wondering if there has been progress on your side regarding plasma sheath.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Sat, Jun 6, 2026 at 8:05 AM

Hi Alireza,

I was able to see the transition from capacitive behavior as we moved the frequencies up. I chose to simulate at higher frequencies to reduce the run time. These code changes have been pushed, I will send you plots for this tomorrow night as I am away this weekend. I feel more confident that the next task of sinusoidal signals would be interesting.

Thank you for the patience

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Sat, Jul 11, 2026 at 12:19 AM

Dear Swadesh,

I was wondering if you could make progress.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Mon, Jul 13, 2026 at 9:08 AM

Hi Alireza,

I have made slow progress. I could see hints of phase transition with frequencies, but they were not convincing. I tried running a bunch of cases at scaled frequencies to view the transition better and align with Tu et al's work. But I do not have any substantial results yet. Apologies for not updating you on the progress and sharing results. I have not been able to prioritize this.
Hope to hear from me this week hopefully with some plots.

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Mon, Jul 13, 2026 at 4:43 AM

Dear Swadesh,
I appreciate your help and look forward to hearing from you.
Regards,
Alireza

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Mon, Aug 10, 2026 at 2:19 AM

Dear Swadesh

I am waiting for your feedback. My student should defend thesis in a few months. I appreciate your help.

Regards,
Alireza

[Quoted text hidden]

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Fri, Aug 14, 2026 at 4:00 PM

Dear Alireza,

I sincerely apologize for the delayed response. I have been investing more time in this project and hoped to send you the final results, but I faced some challenges.
Currently, I am running long simulations and expect to have the results over the weekend. I have also pushed the latest changes to the script to the repository: <https://github.com/sunswadesh/Plasma-Fluid-Simulation-PFFDTD/tree/PffdtSheath>

I have attached a comprehensive progress report detailing everything I have tried so far, including our very early

work. Please note that some sections were generated with AI assistance, so there may be errors. The good news is that we see some positive results of sheath effects. We just need to be rigorous about the physics. I am hopeful that the long runs will be closer to the results of Tu. Ideally, we would like to get some PIC simulation going. Dr. Spencer had a project attempting this, but not sure if he actively worked on it. I will try to revive that project after this.

Is your student planning to use some of this work for her thesis? I am eager to hear more about the thesis project.

Best regards,

Swadesh

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Swadesh



main.pdf
1525K

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Fri, Aug 14, 2026 at 8:09 PM

Dear Swadesh,

I appreciate your email. My interpretation is that you force the plasma sheath and investigate the effects on antenna input impedance. Am I correct?

Is the model able to resolve the sheath itself? I mean if we run the model does the model show the sheath formation?

We are currently investigating radio wave propagation through a plasma slab located at the antenna far field. Attached are some of the early results.

We would like to use sheath simulation results from your work as well. We would also like to include your name in any publications coming out of this research.

Regards,
Alireza

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1089K

swadesh patra <swadeshp@gmail.com>
To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Sat, Aug 15, 2026 at 6:36 AM

Yes, currently the model is just setup for artificial sheath and it's effect on the antenna impedance. Let's get some agreeable results then we can try to make it self consistent.

Swadesh

[Quoted text hidden]

Alireza Mahmoudian <mahmoudian.a@gmail.com>
To: swadesh patra <swadeshp@gmail.com>

Tue, Aug 25, 2026 at 11:43 PM

Dear Swadesh,

You may find the attached paper useful for antenna input impedance in the presence of a plasma sheath

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Journal of Geophysical Research Space Physics - 2007 - Song - High-voltage antenna-plasma interaction in whistler wave.pdf

513K

swadesh patra <swadeshp@gmail.com>

To: Alireza Mahmoudian <mahmoudian.a@gmail.com>

Wed, Aug 26, 2026 at 4:11 PM

Thanks, I'll check it out!

Swadesh

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